Pranav Varadpandey

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.80
St. Joseph's Co-Ed School, Bhopal	Class XII (CBSE)	2020	90.40%
St. Joseph's Co-Ed School, Bhopal	Class X (CBSE)	2018	89.80%

RESEARCH WORK

TiHAN, IIT Hyderabad -- M.Tech Thesis / Autonomous Robotaxi Research

- Developing **real-time adaptive control and autonomous steer-by-wire** systems for an autonomous robotaxi.
- Extending **simulation-validated PSO/ERPSO-PI and finite-enumeration MPC** to real-vehicle validation.
- Developing **autonomous navigation, localization and path-tracking** algorithms.
- Developing **ultrasonic sensor-array fusion** for environment sensing and obstacle detection.
- Implementing vehicle control using **HY-TTC 590, CODESYS and CAN**.
- Hands-on **vehicle mechanical/electrical integration**, including differential axle and VESC.

SKILLS

Programming & DBMS: Python, C, CODESYS, Structured Text (IEC 61131-3), Continuous Function Chart (CFC), MySQL

Control Systems: PI/PID, Model Predictive Control (MPC), Pure Pursuit, Bicycle Model, Drive-by-Wire (DbW)

Autonomous Navigation & Sensing: ROS, Waypoint Navigation, Ultrasonic Sensor Fusion, Obstacle Avoidance

RELEVANT COURSES

Foundations of Machine Learning

Data Science Analysis

Autonomous Navigation

PROJECTS

GNSS-Based Vehicle Navigation and Data Visualization

- Developing an integrated vehicle data pipeline using **ROS, CARLA, Python, and MATLAB/Simulink**, with **SQL** for structured storage and **Power BI** for interactive data visualization.
- Designing a **SQL database** to organize vehicle-state, GNSS, wheel odometry, IMU, and road-condition data for querying, analysis, and performance monitoring.
- Developing **Power BI dashboards** to visualize vehicle trajectories, navigation performance, sensor data, and road conditions, supporting analysis of autonomous navigation and ADAS functions.
- Integrating GNSS-based navigation, Hybrid A* path planning, and PPC/MPC-based path tracking in CARLA to generate vehicle-operation data for visualization and evaluation.

Adaptive Waypoint Navigation & ADAS in CARLA

- Developing a **CARLA-based autonomous navigation and ADAS framework using Python, ROS, MATLAB/Simulink**, beginning with waypoint-based path tracking using the **Pure Pursuit Controller (PPC)**.
- Use of **Gen AI** like LLM APIs for generating **synthetic test cases**.
- Developing **adaptive PPC strategies** to improve path-tracking performance across varying vehicle speeds, with a target operating range extending to **80 km/h**.
- Investigating the robustness of adaptive path tracking under high-speed driving conditions for applications including

obstacle avoidance and overtaking.
- Extending the control framework toward **Model Predictive Control (MPC)** for predictive and constraint-aware vehicle path tracking and for high speed stability.
- Using **LiDAR-derived environmental information** as input to ADAS decision-making for obstacle detection and local driving decisions.

Adaptive Steering Angle Tracking for Steer-by-Wire using Metaheuristic Optimization

- Developed a **real-time low-level steering-angle tracking system in CODESYS** to achieve high-precision steering actuation toward a desired steering angle.
- Implemented and evaluated **adaptive PI control using Particle Swarm Optimization (PSO) and Grey Wolf Optimization (GWO)** for steering-angle tracking.
- Developed a **finite-enumeration Model Predictive Control (MPC)** approach for low-level steering control by evaluating candidate control actions over the prediction horizon.
- Investigated **Entropy-Regularized PSO (ERPSO)** using Shannon entropy to improve swarm exploration and mitigate premature convergence during controller parameter optimization.
- Evaluated closed-loop steering performance using **tracking error, RMSE, MAE and control effort** under vehicle steering trajectories.

POSITIONS OF RESPONSIBILITY

Tinkerer's Lab: **Mentored** students in building a Wall-E-inspired robot, with a focus on **component integration.**

EXTRACURRICULARS

Hobbies : Singing, Yoga, Learning Japanese and Japanese Culture, Anime,

Sports : Squash, Basketball, Tennis